Abu Kaisar Md Faisal

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Education

Rajshahi University of Engineering & Technology (RUET) *Rajshahi, Bangladesh*
Bachelor of Science (B.Sc.) in Mechanical Engineering *Jan 2020 – July 2025*
Cumulative GPA: 3.83/4.00
Relevant Coursework: Aerodynamics, Fluid Mechanics, Engineering Mechanics, Applied Engineering Mathematics, Design of Machine Elements & Machine Tools Design, Solid Mechanics, Computer Aided Design, Heat Transfer.

Research Experience

Advanced Fluid Flow Research Group (AFFRG) | PI: Dr. Mahadi Hasan Masud *Rajshahi, Bangladesh*
Undergraduate Researcher *June 2023 – Present*

- **Thesis:** *Stall-Adaptive Aerofoil: Numerical and Experimental Optimization of Dimpled and Leading-Edge Protuberance Surfaces.*
- **Responsibilities:**
  - Nature-inspired design and biomimetic surface optimization.
  - Numerical aerodynamic analysis, optimization, and experimental wind tunnel testing.
  - Aerodynamic lab development, operation, and maintenance of High-Performance Computing (HPC) cluster.

Research Publications

Peer-Reviewed Journals

- Haque, Md. A., Azad, A. M. A. S., Ankhi, I. J., **Faisal, A. K. M.**, & Masud, M. H. (2026). Survey and analysis of consumer behavior for food waste management in Bangladesh. *Energy Nexus, 22, 100695.* https://doi.org/10.1016/j.nexus.2026.100695
- Saimoon, N. Z., Hasan Masud, M., Salehin, N., Alam, Md. M., **Faisal, A. K. M.**, & Rabiul, M. (2026). Aerodynamic Performance of Bio-inspired Vortex Generator on Automobiles. *Engineering Research Express*. https://doi.org/10.1088/2631-8695/ae3cde
- Masud, M. H., **Faisal, A. K. M.**, Alam, Md. M., Rafi, A. I., Islam, R., Ankhi, I. J., Bai, H., & Wang, H. (2025). Corrugated airfoils for enhanced low-speed aerodynamics: A bio-inspired review. *Physics of Fluids, 37(8)*. https://doi.org/10.1063/5.0278666
- Masud, M. H., Ankhi, I. J., **Faisal, A. K. M.**, & Alam, Md. M. (2025). Translating marine biology into engineering: A review of biomimicry and its applications. *Ocean Engineering, 339*, 122108. https://doi.org/10.1016/j.oceaneng.2025.122108
- Masud, M. H., **Faisal, A. K. M.**, Alam, Md. M., & Dabnichki, P. (2025). Performance analysis of little penguin wing for the application in Micro Aerial Vehicles: A numerical and experimental approach. *International Journal of Aerospace Engineering*. https://doi.org/10.1155/ijae/9989267
- Himel, Md. H. H., Akter, M., **Faisal, A. K. M.**, Ahmed, M. M., & Masud, M. H. (2025). Low-cost innovative food dryers for developing countries: current status and future prospect. *Innovative Food Science & Emerging Technologies, 104*, 104109. https://doi.org/10.1016/j.ifset.2025.104109
- Ankhi, I. J., Hossain, G. A., **Faisal, A. K. M.**, Hasan, Md. R.-U., Barua, S., & Masud, M. H. (2025). Material flow analysis and risk evaluation of informal and formal E-waste recycling processes in Bangladesh: Towards sustainable management strategies. *Journal of Cleaner Production, 497*, 145090. https://doi.org/10.1016/j.jclepro.2025.145090
- **Faisal, A. K. M.**, Ankhi, I. J., Hossain, G. A., Ahmed, M. M., Siddhpura, M., & Masud, M. H. (2025). Emerging concerns associated with E-waste exposure in Bangladesh. *Environmental Science and Pollution Research*. https://doi.org/10.1007/s11356-025-36268-9
- Olayiwola, O. S., Masud, M. H., **Faisal, A. K. M.**, Siddhpura, M., & Siddhpura, A. (2024). Optimization of dimpled surface of NACA0012 airfoil to enhance the aerodynamic performance. *Tuijin Jishu/Journal of Propulsion Technology, 50*. https://doi.org/10.52783/tjjpt.v45.i03.7860

Book Chapters

- **Faisal, A. K. M.**, Ankhi, I. J., & Masud, M. H. (2026). Aerodynamic Improvement of conventional airfoil with biomimetic modification. In *Biomimetics for Aviation and Marine Applications*. Elsevier. *(Invited Book Chapter Under-review)*
- Ankhi, I. J., **Faisal, A. K. M.**, & Masud, M. H. (2026). Impacts of Reynolds number scaling in different stall conditions: A numerical approach. In *Biomimetics for Aviation and Marine Applications*. Elsevier. *(Invited Book Chapter Under-review)*
- **Faisal, A. K. M.**, Ahmed, M. M., Ahmed, R., Ahmed, Md. R., & Masud, M. H. (2025). Innovative methods for transforming food waste into resourceful by-products. In *Resource recycling and management of food waste* (pp. 143–183). Springer Nature Switzerland. https://doi.org/10.1007/978-3-031-86688-3_8
- Ahmed, M. M., Hossain, G. A., **Faisal, A. K. M.**, Ananno, A. A., Rashid, M., & Masud, M. H. (2024). Methanol production from municipal solid waste. In *Reference Module in Chemistry, Molecular Sciences and Chemical Engineering*. Elsevier. https://doi.org/10.1016/B978-0-443-15740-0.00031-8

Conference Papers

- Ahmed, M. M., **Faisal, A. K. M.**, Shahadat, M. Md. Z., Hossain, G. A., & Masud, M. H. (2024, December 19). Numerical investigation of the aerodynamic performance of different airfoil configurations for low-speed vertical axis wind turbine applications. *9th BSME International Conference on Thermal Engineering*.
- Hossain, G. A., **Faisal, A. K. M.**, Ankhi, I. J., Dabnichki, P., & Masud, M. H. (2024, December 19). Design modification of Slocum underwater glider: A numerical approach. *9th BSME International Conference on Thermal Engineering*.
- **Faisal, A. K. M.**, Ankhi, I. J., Hossain, G. A., & Masud, M. H. (2024, December 12). Blending airfoils: An innovative approach to improve the aerodynamic performance of a vertical axis wind turbine. *6th International Conference on Mechanical, Industrial and Materials Engineering*.

Teaching Experience

Khwaja Yunus Ali University (KYAU) *Sirajganj, Bangladesh*
Lecturer *August 2025 – Present*

- Delivered lectures, tutorials, and laboratory sessions for undergraduate students in Mechanical Engineering.
- Supervised students in production process optimization, CAD, and Finite Element Method courses.
- Developed course materials integrating computational simulation and design-based learning.

Industrial Training

- Completed a 10-day training at the **Bangladesh Power Development Board (BPDB)**, investigating power plant operations and fluid flow systems.
- Conducted industrial visits to **Barapukuria Thermal Power Station**, **Walton**, and **Katakhali Power Plant**, assessing turbine components, aerodynamic design features, and manufacturing processes.

Skills

- **Technical Tools:** SolidWorks, ANSYS, AutoCAD, MATLAB, Python, C++, ModeFrontier, Adobe Suite, Minitab, MS Office, Cura Slicer.
- **Engineering:** Mechanical Design (SolidWorks, AutoCAD, Fusion 360), Additive Manufacturing (3D Printing with PLA, SLA, and FDM), Computational Fluid Dynamics (Fluent), Finite Element Analysis (Abaqus, ANSYS), Optimization (multi-objective and surrogate modeling in Python & MATLAB).
- **Languages:** English: IELTS — Band 7.0 (L: 7.5, R: 7.5, W: 6.5, S: 6.0), Bengali (Native).

Projects

- **IoT-based Rechargeable Modules with Temperature, Humidity, CO_2, and O_2 Sensors.** *September 2025*
- **Real-Time Thermal Mapping of Density-Driven Fluid Convection in Water via ESP32** *May 2024*
- **Automated Liquid Level and Thermal Regulation System via Arduino UNO** *Mar 2023 – Aug 2023*
- **PET Filament from Plastic Bottles for 3D Printing** *June 2023*

Extra-Curricular Activities

Society of Computer Aided Designers (SCADR) *RUET, Rajshahi*
General Secretary *May 2024 – July 2025*

- Organized CAD & 3D-Printing workshops for 180+ participants; trained 120+ students in SolidWorks and ANSYS, improving design proficiency by 30%.
- Conducted a 10-day SolidWorks training collaboration with BAIUST Robotics Society under "Empowering with SolidWorks."

Society of Automotive Engineers (SAE-RUET) *RUET, Rajshahi*
Joint Secretary *June 2024 – July 2025*
- Organized the seminar *"Navigating the Automobile Engineering Landscape"* for 120+ attendees.
- Led the aerodynamic optimization of a solar car for the Bharat Solar Vehicle Challenge using CFD; instructed 40+ members in automotive aerodynamics and spoiler design analysis.

Robotic Society of RUET (RSR) *RUET, Rajshahi*
Vice President *July 2024 – July 2025*
- Directed "Techmayhem 2023" with 150+ participants; mentored 30+ juniors in Python for robotics.
- Led the development of a sensor-integrated robotic arm improving operational precision.

Team Ogrodoot (RSR) *RUET, Rajshahi*
Mechanical Designer *Dec 2022 – Dec 2024*
- Designed suspension and wheel components for a Mars rover, achieving 11th place globally in the *2023 International Rover Design Challenge*.
- Reduced CAD design iteration time by 20% using efficient parametric modeling workflows in SolidWorks.

10 Minute School *Rajshahi, Bangladesh*
Content Designer *Feb 2022 – Dec 2023*
- Designed PowerPoint-based educational content for 10,000+ users.
- Managed project timelines and boosted learner engagement through improved visual content strategy.

Awards and Achievements

- 1st Place — Best Research Publication Award (Faculty of Mechanical Engineering, RUET, 2025)
- 1st Place — Best Research Publication Award (Department of Mechanical Engineering, RUET, 2025)
- Merit-based Scholarships, RUET (2021–2023)
- 2nd Place — Techmayhem CAD Design Competition (2023)
- 11th Place — International Rover Design Challenge (2023)
- Technical Scholarship — RUET (2020)
- College Prefect (2018); Excellent Discipline Academic Award (2018)
- Education Board Scholarship (10^{th} Grade, 2017)
- 1st Place (Regional) — National High School Programming Competition (2017)
- 2nd Place (Regional) — National High School Programming Competition (2016)

Certifications

- Certified SOLIDWORKS Professional (CSWP) — Mechanical Design
- Certified SOLIDWORKS Associate (CSWA) — Mechanical Design, Electrical, Additive Manufacturing, Sustainability
- Certified 3DSwymer Associate
- Python for Everybody Specialization (Coursera)
- Python 3 Programming Specialization (Coursera)
- Introduction to Scripting in Python (Coursera)
- Google IT Automation with Python Specialization
- Autodesk CAD/CAM/CAE for Mechanical Engineering
- Excel Skills for Business: Essentials
- Neural Networks and Deep Learning
- Computational Fluid Mechanics - Airflow Around a Spoiler
- FEM - Linear, Nonlinear Analysis & Post-Processing
- CFD Simulation Through a Centrifugal Pump
- Finite Element Analysis Convergence and Mesh Independence
- Mastering Unsteady CFD of Vertical Axis Wind Turbine